

2022 9th International Conference on Power and Energy Systems Engineering (CPESE 2022),
Doshisha University, Kyoto, Japan, 9–11 September 2022

JEPX spot prices forecasting system using GIS

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Received 24 October 2022; accepted 25 October 2022

Available online 10 November 2022

Abstract

Japan Electric Power Exchange (JEPX) is the only electricity market in Japan that allows transactions for electric power established in 2005. Several commodities are traded there, such as the spot market and the forward market. In particular, the spot market is a major trading market. The price of the spot market changes depending on the relationship between supply and demand. Therefore, it is important to forecast spot market prices to make a supply and demand plan for the next day. This research focused on the factors that determine supply and demand are related to geospatial information—first, deriving the explanatory variables for JEPX spot prices using Geographic Information System (GIS). Then, constructing the spot price forecasting system by machine learning using the derived explanatory variables. By using this system, it is possible to forecast electricity prices with higher accuracy than existing methods.

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Peer-review under responsibility of the scientific committee of the 9th International Conference on Power and Energy Systems Engineering (CPESE 2022).

Keywords: Japan electric power exchange; Geographic information system; Spot price; Meteorological data

1. Introduction

In recent years, the total trading volume of the JEPX is continuing to increase, and the JEPX trading volume accounts for 41.5% of Japan's electricity demand as of December 2021 [1]. JEPX has the spot market for trading electricity the next day and the forward market for trading electricity few days or years ahead. In particular, the spot market is a major trading market of the JEPX, and 48 products are traded every 30 min, 24 h a day. The electric utilities must bid by 10 AM the day before the trading day, and the relationship between supply and demand determines the transaction price. In general, transaction prices are determined by a complex relationship between various factors related to the supply and demand of electric power. As a trend in recent years, the electric power market price is strongly fluctuating due to the increase in the ratio of renewable energy such as solar power generation. It has been reported that the overseas electric power market price is also strongly fluctuating due to the increase in the ratio of renewable energy [2]. In the Japanese electricity market, the amount of renewable energy is

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expected to continue to increase in the power system based on the 6th Strategic Energy Plan, so it is expected that the electric power market price will strongly fluctuate in the future [3].

The relationship between supply and demand of electric power changes depending on many factors, such as weather conditions, fuel prices, and power plant shutdown information. Several methods have been proposed for forecasting electricity prices using this information. Sato et al. have been proposed the price forecasting method using explanatory variables considering the correlation between price and temperature [4]. Since there is a strong correlation between temperatures and electricity prices, it is possible to forecast the prices with higher accuracy than conventional methods. Manaka et al. have been proposed the price forecasting method that integrates multiple forecasting methods [5]. This method shows that forecast errors can be reduced by integrating multiple forecasting methods. Kikuchi et al. have been proposed the price forecasting method that using simplified merit order model [6]. Furthermore, there are multiple price forecasting methods in overseas electricity markets [7]. These methods use the weather forecast information as explanatory variables. In addition, these explanatory variables become more closely related to electricity prices by relating to geographical information. However, the construction of the electric power market price forecasting model focusing on geographic information have not been done. The purpose of this study is to reduce the power purchase cost of power consumers by constructing the spot prices forecasting system with highly forecast accuracy using geographic information. In this paper, the authors derived the population-weighted temperature data and the forecasted amount of solar power generation data, closely related to the spot market prices in the Kyushu area, the western area of Japan, using the Geographic Information System (GIS). Furthermore, the authors constructed the spot market prices forecasting model with a highly forecast accuracy by combining these data and other explanatory variables in the Kyushu area.

2. The overview of GIS and used meteorological data

2.1. GIS utilization overview

GIS stands for Geographic Information System, and it is a technology for modeling the real-world infrastructure on a computer. GIS can deal with several data based on Geographic Information, and GIS can perform several analyses by using these data. GIS uses layer data (e.g., land use data, elevation data) for modeling the real world. Therefore, GIS can deal with several data at once, and it is possible to support the user's decision. The overview of the main functions and data of GIS is shown in Fig. 1.

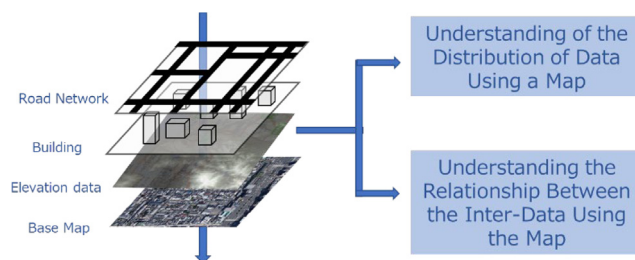


Fig. 1. The basic principle of GIS.

GIS can also handle weather information. For example, GIS can represent the time change of meteorological information on a map by preparing a layer of surface meteorological data for each time. In this paper, the authors used Inverse Distance Weighted (IDW) [8], a method of complementing point data and spatial join [9], which combines information between superimposed layers, which are functions of GIS.

2.2. Used meteorological data

Since the spot market price is related to the meteorological data, the authors derived the explanatory variables based on the data. In this study, the authors used the numerical weather prediction Grid Point Value (GPV), which is the weather forecast data provided by the Japan Meteorological Agency. The numerical weather prediction GPV is a numerical weather prediction model that provides a weather forecast by gridding entire earth's atmosphere with

a 20 km mesh in the horizontal direction and 60 layers in the vertical direction. Numerical weather prediction GPV can forecast the weather up to 11 days later, the weather forecast up to 132 h later is calculated four times a day, and the weather forecast up to 11 days later is calculated twice a day. Weather forecast data is provided as hourly data. The authors derived explanatory variables using the temperature forecast and solar radiation forecast data provided by numerical weather prediction GPV. The authors used DeGRIB to convert these weather forecast data into a format that can be used on GIS [10]. DeGRIB is software that can convert weather forecast data provided in Grib2 format into shapefiles that can be handled by GIS. The weather forecast data can be acquired as point data of points on the grid. Fig. 2 shows a map with weather forecast point data on GIS. The figure shows that the information is represented on the map at regular intervals as point data. In addition, as shown in Fig. 3, each point contains latitude and longitude information and temperature forecast data at that point. Solar radiation forecast data also contains these data.

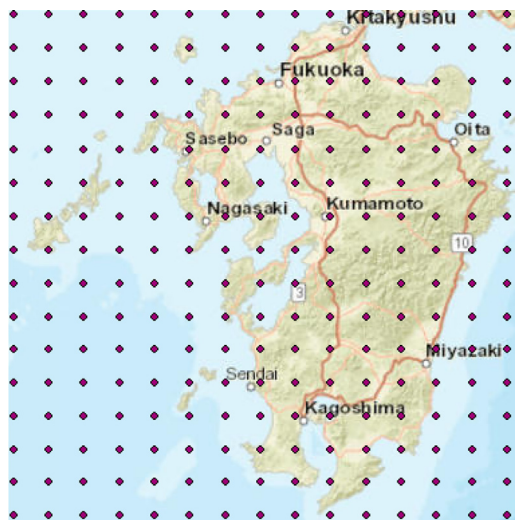


Fig. 2. Weather forecast data on GIS.

FID	Shape	POINTID	X	Y	Lon	Lat	TMP
0	Point	10001	1	1	120	20	25
1	Point	20001	2	1	120.25	20	24
2	Point	30001	3	1	120.5	20	24
3	Point	40001	4	1	120.75	20	25
4	Point	50001	5	1	121	20	25
5	Point	60001	6	1	121.25	20	25
6	Point	70001	7	1	121.5	20	25
7	Point	80001	8	1	121.75	20	25
8	Point	90001	9	1	122	20	25
9	Point	100001	10	1	122.25	20	25
10	Point	110001	11	1	122.5	20	25
11	Point	120001	12	1	122.75	20	25
12	Point	130001	13	1	123	20	25
13	Point	140001	14	1	123.25	20	25
14	Point	150001	15	1	123.5	20	24
15	Point	160001	16	1	123.75	20	24
16	Point	170001	17	1	124	20	24
17	Point	180001	18	1	124.25	20	24

Fig. 3. Data contained in the weather forecast data.

3. Derivation of JEPX spot price explanatory variables using GIS

3.1. Population-weighted temperature

The relationship between supply and demand determines the spot price, and one of the factors that determine the electricity demand is temperature. Mainly in summer and winter, the power demand changes depending on the power consumption of the air conditioner, especially in the morning and evening. For example, on cold days,

the power consumption of the heating mode increases; therefore, the demand for electric power increases. For this reason, the spot price is related to the temperature, and the price tends to rise suddenly, especially on hot days in the summer and on cold days in the winter. In addition, the power consumption of air conditioners relates to the population. For example, if the power consumption of air conditioners increases due to the influence of temperature in a densely populated area, the spot price will rise. For the mentioned reasons above, the authors superimpose the population distribution and temperature forecast on GIS to derive population-weighted temperature data. Furthermore, the authors used it as an explanatory variable to determine the spot price.

As shown in Fig. 2, the temperature forecast data is given as point data, but it is impossible to acquire the temperature between points with this data. Therefore, the authors used IDW, a GIS function, to interpolate between points. IDW is a method of interpolation by weighting inversely proportional to the distance. IDW makes it possible to create spatially continuous temperature data.

Next, the authors superposed temperature data and population data. Fig. 4 shows the superposition of forecast temperature data after IDW and population data on GIS. The dark areas indicate low temperatures, and light colors indicate high temperatures. The authors used the 2015 census population data by the municipality as population data on GIS. In addition, since the population changes depending on the time zone, the temperature data is weighted using the daytime population from 7 AM to 8 PM, and the temperature data is weighted using the nighttime population during other time periods. The authors used spatial join, a GIS function, to weight temperature data. The authors calculated the forecast temperature for each municipality by using spatial join. Finally, the authors calculated the population-weighted temperature data for the entire Kyushu area every 30 min using (1). In (1), T_p is the temperature data weighted by population, P_n is the population of the n th municipality, T_n is the forecasted temperature of the n th municipality, and P_k is the population of the entire Kyushu. In this paper, the population-weighted temperature data are the same in two consecutive time periods since the authors used hourly weather forecast data.

$$T_p = \frac{\sum(P_n \times T_n)}{P_k} \quad (1)$$

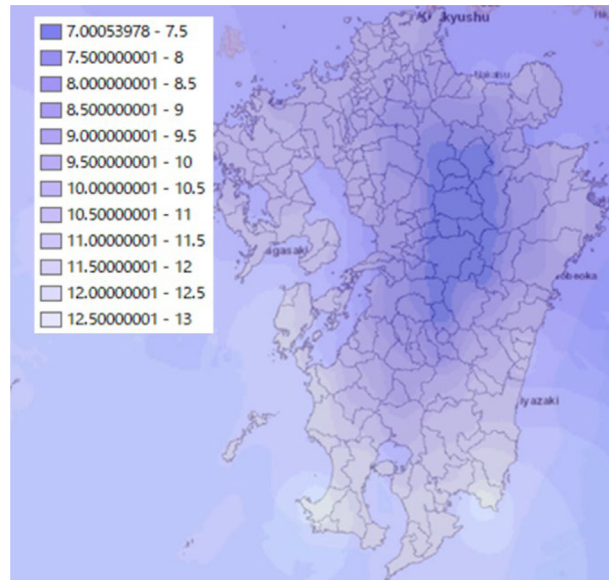


Fig. 4. The superposition of forecast temperature data and population data on GIS. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Forecasted amount of solar power generation

In recent years, the amount of solar power generation has been increasing, and the spot price has changed dramatically depending on the amount of solar power generation. Especially in the Kyushu area, the ratio of solar

power generation to the total amount of electricity generated is larger than in other areas. Therefore, the amount of solar power generation has a large effect on the spot price. Furthermore, the amount of power generated by solar power generation varies depending on the amount of solar radiation in the place where solar power generation is installed. Therefore, the amount of solar power generation is related to geographic information. For these reasons, the forecasted amount of solar power generation data can be derived by superimposing the distribution of photovoltaic power generation equipment and the forecasted solar radiation on GIS and used it as an explanatory variable to determine the spot price.

The authors interpolated the forecasted solar radiation point data with IDW and obtained the forecasted solar radiation data for each municipality by spatial join. Fig. 5 shows the superposition of forecast solar radiation data after IDW and solar power generation introduction data on GIS.

After that, the authors calculated the forecasted solar power generation in the entire Kyushu area every 30 min using (2). In (2), Ps is the forecasted amount of solar power generation, Gn is the amount of solar power generation introduced by the n th municipality, In is the forecasted solar radiation of the n th municipality, and S is the standard solar radiation indicates strength.

$$Ps = \sum (Gn \times In \div S) \quad (2)$$

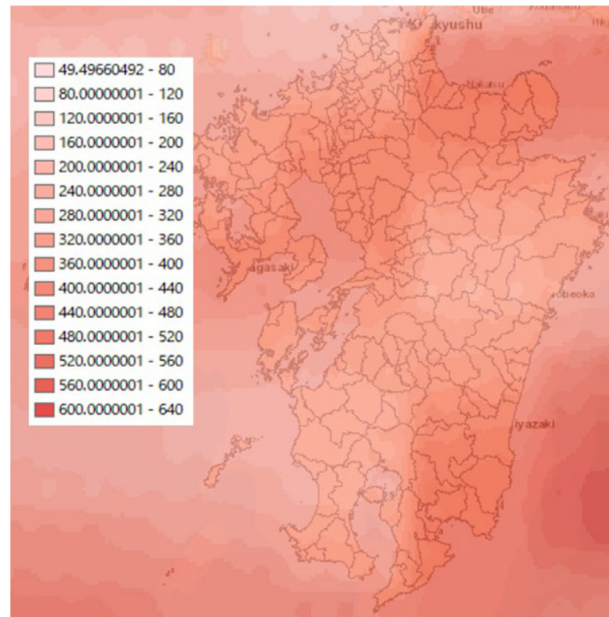


Fig. 5. The superposition of forecast solar radiation data and solar power generation introduction data on GIS.

4. Construction of the JEPX spot price forecast system and evaluation of accuracy

4.1. Simulation conditions

The authors constructed a prices forecasting model using the derived population-weighted temperature data and forecasted amount of solar power generation data as explanatory variables. In general, electric power demand varies greatly between weekdays and holidays. Therefore, the authors used a dummy variable for weekdays and holidays, and a dummy variable for days of the week as explanatory variables. In addition, the authors used the power plant shutdown information published by Hatsuensho Joho Kokai System (HJKS) as an explanatory variable to represent the supply [11]. The deadline for bidding on the JEPX spot market is 10 AM the day before the transaction date, however the authors built the model using the data as of 3 PM two days before the transaction date to assume the actual use.

The authors used Random Forest, one of the machine learning methods, to build the spot prices forecasting model. A random forest is an ensemble model that combines multiple prediction models called “decision trees” and it is a supervised machine learning algorithm. The random forest provides the average forecasted value of each decision tree as the forecasted value. There is a method using the random forest in previous research [5].

The authors constructed a spot prices forecasting model and performed prices forecasting simulation from April 2019 to December 2021. The authors predicted the spot price by using the data from January 2018 to the day before the forecast target day as learning data and using each explanatory variable of the forecast target day as an input. The authors evaluated the accuracy of price forecast by Mean Absolute Error (MAE) shown in (3). In (3), N is the number of data, $\hat{y}$ is the forecasted price and y is the actual spot price.

$$MAE = \frac{\sum_{n=1}^N |\hat{y} - y|}{N} \quad (3)$$

4.2. JEPX spot price forecasting and evaluation

The authors forecasted the spot price in the Kyushu area under the simulation conditions shown above. Table 1 shows the forecast results for Fiscal Year (FY) 2019, FY2020, and FY2021 (until December). The forecast results for FY2019 and FY2020 also include the prediction accuracy of Omura et al.’s model for comparison [5]. However, JEPX spot prices rose to an unprecedented level from December 2020 to January 2021. Since it is difficult to forecast using the proposed method during this period, the authors excluded the data from December 15, 2020, to January 31, 2021, from the forecast targets. These periods are similarly excluded in previous research. In addition, the authors also exclude this period from the training data

Table 1. The comparison of forecast accuracy.

Fiscal year	Constructed model MAE (yen/kWh)	Omura et al.’s model MAE (yen/kWh)
2019	1.07	1.20
2020 (excluded the prices rose period)	1.13	1.53
2021 (Until December)	1.94	—

As shown in Table 1, it can be seen that good forecast results were obtained even when compared with the existing method. In addition, although the forecast target year is different, it is possible to forecast the price more accurately than the existing method [12]. Fig. 6 shows a comparison between the forecast results and the actual prices for March 23, 2021. As shown in the figure, it can be seen that the price can be forecasted accurately even on a day when there is a price difference between the highest price and the lowest price.

In Table 1, the forecast accuracy for FY2021 is low compared to other years. Fig. 7 shows the monthly average of the forecast errors from April 2021 to December 2021. The data shows the forecast accuracy deteriorates from October to December. During this period, the spot price of JEPX rose. Since the price rise in the low demand period in autumn is unprecedented in the past, the authors consider that the forecast accuracy has deteriorated. Nevertheless, the price forecast is possible with the same forecast accuracy as in other years, excluding this price rise period from Fig. 7.

5. Conclusion

This study focused on JEPX spot prices related to geographic information. The authors derived the population-weighted temperature data as a factor that determines the power demand, and the forecasted amount of solar power generation data as a factor that determines the power supply by using GIS. The authors constructed a JEPX spot prices forecasting model using these data and other open data based on machine learning. The authors performed a prices forecasting simulation and showed prosperity forecasting results using a constructed model compared to the existing method. It is possible to reduce electricity costs for electric consumers by combining this forecast model with storage batteries and EVs and performing energy management. In the future, the authors will consider the factors of error and work on further improving forecast accuracy.



Fig. 6. A comparison between the forecast results and the actual prices for March 23, 2021.

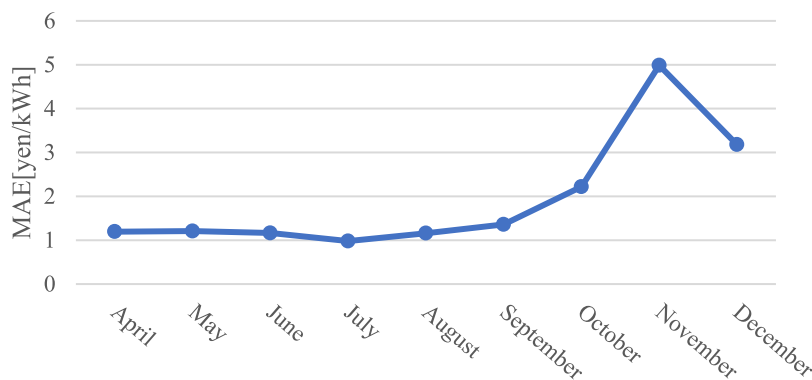


Fig. 7. The forecast accuracy for FY2021.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was supported by Kyutech Research Fellowship. I would like to thank to the research members for their kind assistance.

References

- [1] Documents of the 71st institutional design expert meeting, Electricity and Gas Market Surveillance Commission, 2022, p. 3.
- [2] I. Milstein, A. Tishler, Can price volatility enhance market power? The case of renewable technologies in competitive electricity markets, *Resour Energy Econ* 41 (2015) 70–90.
- [3] The 6th strategic energy plan, Agency for Natural Resources and Energy, 2022, https://www.enecho.meti.go.jp/en/category/others/basic_plan/.

- [4] Yuki. Sato, et al., Relationship between JEPX spot price prediction accuracy and temperature by SARIMAX model, in: 2016 Tohoku-section joint convention of institute of electrical and information engineers, Japan, 2D07, 2014.
- [5] Manaka. Omura, et al., Forecast of area prices in Japanese day-ahead electricity market using open data -examination of approach to combine multiple forecasting schemes, *IEEJ Trans Power Energy* 141 (5) (2021) 366–373.
- [6] Kota. Kikuchi, Shinji. Wakao, Kosuke. Harada, Yuko. Hoshino, Versatility verification of JEPX spot price prediction with simplified merit order model, in: Proceedings of the 2022 annual meeting IEE of Japan, 2022, pp. 215–216.
- [7] R. Weron, Electricity price forecasting: A review of the state-of-the-art with a look into the future, *Int J Forecast* 30 (4) (2014) 1030–1081.
- [8] Esri, IDW (Spatial Analyst), <https://pro.arcgis.com/en/pro-app/2.8/tool-reference/spatial-analyst/idw.htm>.
- [9] Esri, Spatial Join (Analysis), <https://pro.arcgis.com/en/pro-app/2.8/tool-reference/analysis/spatial-join.htm>.
- [10] Tutorial: Using NDFD gridded data in a GIS environment, The Metrological Development Laboratory, <https://vlab.noaa.gov/web/mdl/ndfd-gis-tutorial>.
- [11] Power generation information disclosure system, Japan Electricity Power Exchange, <https://hjks.jepx.or.jp/hjks/top>.
- [12] Soichiro. Moridaira, Electric prices forecasting by temperature options, *J Real Options Strategy* 11 (1) (2020) 35–45.